

Project Testing

The application was tested using the **Pytest** framework to verify the correctness of individual service modules and API endpoints. FastAPI's **TestClient** was used to perform API testing without deploying the application on a live server. Each core functionality was tested independently to ensure reliable integration between the frontend and backend.

Test Files

Test File	Description
<code>test_analyze_event.py</code>	Tests the DistilBERT-based event theme extraction service.
<code>test_topic_generator.py</code>	Tests the Gemini conversation generation module.
<code>test_fact_check.py</code>	Tests the Wikipedia fact-checking service.
<code>test_routes.py</code>	Tests FastAPI API endpoints and request/response validation.

Code Coverage: Automated testing was performed using **Pytest** with the **pytest-cov** plugin. The application achieved an overall **73% code coverage**, with complete coverage (100%) for the Event Analyzer, Topic Generator, and Schema modules. Core API routes and the Fact Checker also achieved high coverage, demonstrating that the major functionalities of the application were successfully validated through automated tests.